ANTI-SLEEP ALARM SYSTEM FOR DRIVERS

Mini Project 1 B Report

Submitted in partial fulfillment of the requirement of University of Mumbai

For the Degree of

(Electronics & Telecommunication Engineering)

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CERTIFICATE

This is to certify that the mini project 1 A entitles "**ANTI-SLEEP ALARM SYSTEM FOR DRIVERS**" is a bonafide work of

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submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the Bachelor of Engineering (Electronics and Telecommunication Engineering).

Prof. Vijaypal Yadav		
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Project Report Approval

This Mini Project 1 A Report – entitled "**Anti-Sleep Alarm System for Drivers**" by following students is approved for the degree of ***S.E. in "Electronics and Telecommunications Engineering"***.

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Examiners Name & Signature:

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Date: 20/4/25

Place: ----------

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

With the predictions of World Health Organization (WHO) that the number of deaths due to traffic accidents will be around 2 million within less than 15 years, researchers nowadays are paying more attention to how to help in preventing traffic accidents and lower the number of occurred fatalities. The purpose of this study is an attempt to prevent traffic accidents due to fatigue or sleepiness. In this report, a portable and low cost device for prevention of accidents that happen because of sleepiness or fatigue. The proposed system consists of two main parts that detect eye blinking based on IR sensors mounted on eyewear. Depending on the reflected and absorbed IR radiation, this system detects and classifies the eye blinking into normal blinking (NB).

Chapter 1

Introduction

Everyone knows about the alarms that abruptly wake us from our slumbers each morning, but have you heard of alarms that can keep us awake while we're driving? Road traffic injuries and deaths have a terrible impact on individuals, communities and countries. Drowsy driving is defined as operation of a motor vehicle while being cognitively impaired by lack of sleep. According to the National Sleep Foundation, some of the drowsy driving signs are difficulty focusing, yawning repeatedly or rubbing eyes and trouble keeping head up. In recent years, driver drowsiness has been one of the major causes of road accidents and can lead to severe physical injuries, deaths and economic losses. Annually and worldwide, over 1.3 million people die each year on the road and 20-50 million people suffer non-fatal injuries due to road accidents. According to National Sleep Foundation surveys, half of American adults consistently report that they have driven drowsy and 20% admit that they have actually fallen asleep at the wheel in the previous year. These statistics suggest that driver drowsiness is one of the most dominant causes for car accidents, injuries, deaths and economic losses. Hence, developing a system for monitoring drivers' drowsiness and alerting the driver when he/she is not paying enough attention to the road is a fundamental way to prevent accidents to prevent traffic accidents caused by sleepy drivers, many systems have been proposed. In this report we will study that with the help of a few components we have made an anti sleep alarm. Here we use components like relay, piezo buzzer, battery, etc. When he/she falls asleep while driving then due to the piezo buzzer it will alert and the driver will wake up.

Chapter 2

Problem Statement

2.1 Problem statement

Develop an anti-sleep alarm system for drivers with an additional engine cutoff feature to enhance road safety. Driving while fatigued or drowsy poses a significant risk to road safety, as it increases the likelihood of accidents due to delayed reactions and impaired judgment. To address this issue, there is a need for an advanced anti-sleep alarm system for drivers. The system should effectively detect signs of drowsiness or fatigue in real-time and alert the driver, preventing potential accidents. However, in addition to the alert mechanism, there is also a requirement for an engine cutoff feature. In situations where the driver fails to respond to the alarm and continues to exhibit signs of extreme fatigue or sleepiness, it becomes necessary to automatically disable the engine, bringing the vehicle to a safe stop. This feature ensures that the driver's safety is prioritized, as well as the safety of other road users. The system should be capable of detecting various indicators of fatigue, including but not limited to eye closure, head nods, erratic steering, and changes in vehicle speed or lane position. Upon detection of potential drowsiness, the alarm should activate auditory, visual, and/or haptic alerts to awaken the driver and regain focus on the road To ensure seamless integration and compatibility with different vehicle models, the system should be designed to interface with the vehicle's electrical system and provide an engine cutoff mechanism. The engine cutoff should be implemented as a fail-safe measure, allowing the driver a short response time too the system in case of false positives or emergencies

1.1 Features

1. Real-time eye monitoring
2. Nodding detection
3. Customizable settings to alert drivers of drowsiness

1.2 Objectives

1. Early Detection of Drowsiness
2. Prompt Alert Mechanism
3. Customizable Sensitivity Levels
4. Engine Cutoff Mechanism
5. Fail-Safe Override Capability

Chapter 3

Literature Survey

Auto-driver drowsiness can be detected using artificial intelligence and visual information. The purp ose of the system detect, track and study the faces and eyes of the drivers to make a real vehicle ima ge different from the drivers to validate the algorithms. It is a real-time system that works in differen t lighting conditions. The frequency of accidents has increased due to several factors, one of is also i mplemented with a video-based approach.

This system is non-invasive and uses human-related elements. Band power and empirical state deco mposition methods are used to study and extract the signal, SVM (Support Vector Machine) is used to confirm the analysis and classify the driver's vitality. The system intends to search for driver drow siness using the hypothesis of Bayesian networks.

The interaction between driver and vehicle characteristics is extracted to obtain reliable driver drows iness symptoms. It provides more appropriate and accurate strategies for designing a drowsy driver detection system. The sleepiness detection system uses brain and visual activity.

The EEG-based sleep detector uses the electroencephalographic (EEG) channel to monitor brain acti vity, diagnostic techniques and fuzzy logic. Use of eyeblink detection and characterization in visual activity monitoring. The electrooculography (EOG) channel is used to extract the blink functions. I mage processing and pattern classification are used to photograph the driver's face, track the driver's facial features, and classify the driver's sleepiness.

After studying the activity of the facial muscles with the Active Appearance Model (AAM), 17 diffe rent characteristic points are determined. The k-nearest neighbour method classifies sleepiness into 6 levels, this method also detected the driver's smile. The head position estimation method is used to d etect the deviation of the tired driver. In this method, Viola and Jones' algorithm for driver Tace d etection.

Design and Implementation

1. Circuit Diagram

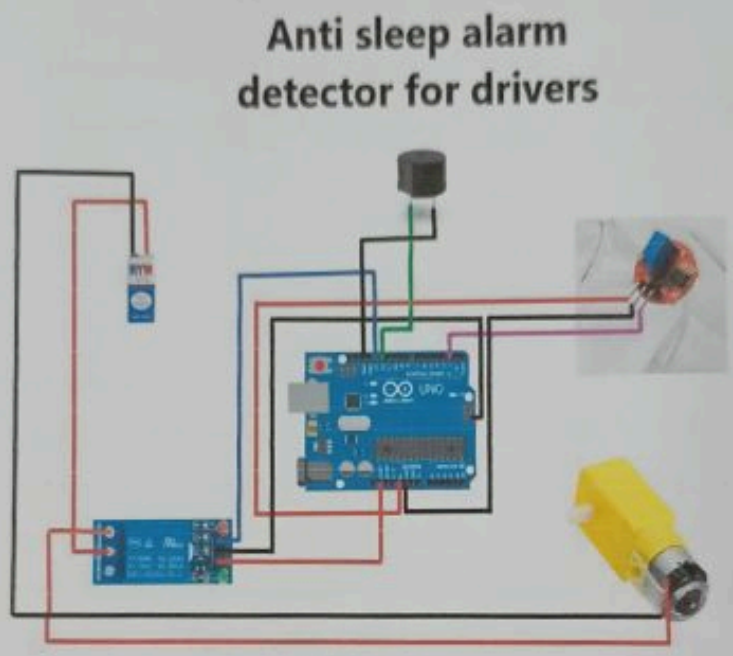

Fig no 1

2. Components Used

- Arduino Uno
- Single Channel 5V Relay
- Eye blink sensor
- Gear Motor
- Buzzer
- 9V Battery

ARDUINO UNO:

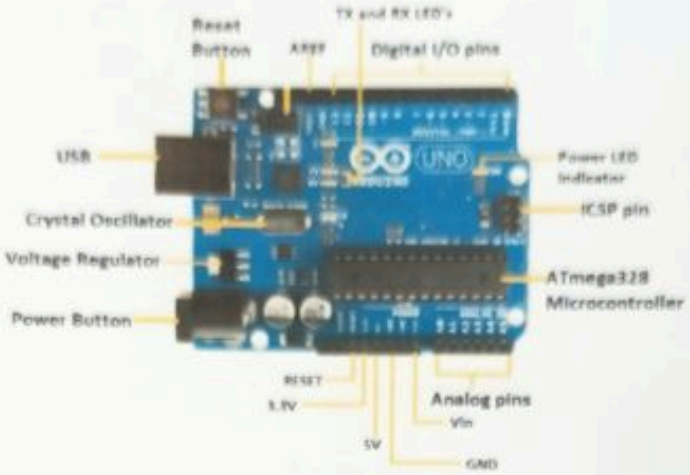

Fig no 2

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

SINGLE CHANNEL 5V RELAY:

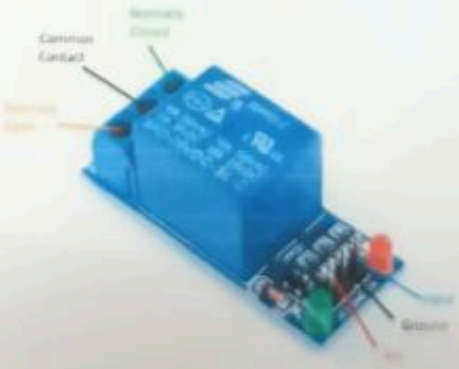

Fig no. 3

Relay is one kind of electro-mechanical component that functions as a switch. The relay coil is energized by DC so that contact switches can be opened or closed. A single channel 5V relay module generally includes a coil, and two contacts like normally open (NO) and normally closed (NC). This article discusses an overview of the 5V relay module & its working but before going to discuss what is relay module is, first we have to know what is relay and its pin configuration.

EYE BLINK SENSOR:

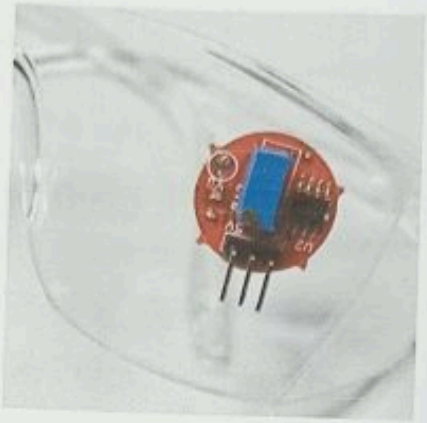

Fig no. 4

The main component of the Eye Blink Sensor is the infrared sensor. The infrared sensor contains two parts. An IR transmitter and an IR receiver. The IR transmitter emits infrared waves onto the eye. While the receiver continuously searches for variations in the reflected waves which indicates that the eye has blinked or not. If the eye is closed then reflected waves will be received by the IR Receiver and the output of the sensor would be low. If the eye is open then all the IR rays will be absorbed by the eye and the output of the eye blink sensor would be high.

BUZZER:

Fig no. 5

The buzzer is a sounding device that can convert audio signals into sound signals. It is usually powered by DC voltage. It is widely used in alarms, computers, printers and other electronic products as sound devices. It is mainly divided into piezoelectric buzzer and electromagnetic buzzer. According to different designs and uses, the buzzer can emit various sounds such as music, siren, buzzer, alarm, and electric bell.

9V BATTERY:

Fig no.6

The 9V Battery is a common small battery widely used in various portable electronic devices. It is known for its reliability and portability, and is an ideal power source for many small devices. This article will introduce the definition, types, uses and application scenarios of the 9V battery in detail.

JUMPER WIRES:

Fig no. 7

Jump wires, also known as DuPont wires, are electrical wires with connectors on each end that enable the creation of circuits without soldering. They are frequently used on breadboards, which have slots specifically designed to receive these connectors.

ARDUINO PROGRAMMING CABLE:

Fig no. 8

An Arduino programming cable connects your computer to your Arduino board, letting you upload code and power your projects. It's a USB cable with a standard Type-A connector for your computer and a Type-B (or C on newer IJSART - Volume 10 Issue 3 – MARCH 2024 ISSN [ONLINE]: 2395-1052 Page | 137 www.ijsart.com boards) connector for your Arduino. While any matching USB cable technically works, consider cable length, quality, and features when choosing one. You can find them online or at electronics stores.

3. Working of system

To start the circuit we first click the 'ON' button on the SPST switch. Then the electrical supply will pass and the motor will start working because there was a loop of 9V battery. The relay will remain closed and the circuit will open. Through an IR sensor we will detect the output. Output we will get with the help of Arduino UNO. While driving if the eye will close for 5 sec then the buzzer will detect. After buzzing for 2 sec then also the eye will close so the relay will open and the circuit will close. That means the car will stop.

Chapter 5

Results

Fig no. 9 **This is the prototype**

An Anti-Sleep Alarm System for Drivers is a safety device designed to detect signs of drowsiness or fatigue in drivers and alert them before they fall asleep at the wheel. It typically uses sensors to monitor indicators like:

- Eye blink rate or eye closure (via a camera or IR sensor)
- Head movement or nodding
- Steering behavior or driving patterns

When the system detects that the driver is getting drowsy, it triggers an alert — such as a buzzer, vibration, or voice message — to wake the driver up and prevent accidents.

It's especially useful for long-distance truck drivers or anyone who drives for extended periods. Let me know if you want help explaining how your version works!

Conclusion and Future Scope

1. Conclusion

In this project a complete driver alert system was implemented. The system is operated by a 9V DC battery. The device is small in size, has a low weight and it is user-friendly and reliable. The device detects the driver's prolonged eye blinking. Consequently, it alarm the driver and woke her/him up and stopped the car. The performance of the system can be enhanced by a smaller IR sensor with lower heating effect. This research achieves good results for the detection and analysis of prolonged eye blinking and may become a universal technique for human life saving in the future

Future Scope

A project to build an anti-sleep alarm system using Arduino has great potential to improve driver safety. Here's a breakdown of the scope:

Main functionalities:

- **Drowsiness detection:** This will be the core function achieved by the eye sensor mounted on the driver's spectacles. The sensor will monitor eye blinks and detect extended closure (e.g., exceeding 3 seconds) as an indicator of drowsiness.
- **Alarm triggering:** When drowsiness is detected, the system will trigger an alarm to alert the driver. This could involve sounds (buzzer), vibrations (motor), or lights (LEDs).
- **Vehicle deactivation (optional):** This is a more advanced feature that involves interfacing with the vehicle's control systems. If the driver remains unresponsive to the alarm after a set time, the system could initiate actions like slowing down, activating hazard lights, or even coming to a controlled stop.

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